

Bangalore Real Estate Price Prediction & Valuation Engine

An Econometric & Machine Learning Valuation Study | Author: Abhiram

1. Executive Summary & Market Dynamics

Bangalore (Bengaluru), India's Silicon Valley, represents one of Asia's most dynamic residential real estate markets. Driven by rapid IT corridor expansion, property prices display extreme micro-market variance: central heritage localities (Indiranagar, Rajaji Nagar) command **₹12,000–₹18,000+ per sq ft**, whereas peripheral tech clusters (Electronic City, Sarjapur) trade between **₹3,500–₹5,500 per sq ft**. Public listing aggregators suffer from data anomalies, unstandardized square footage ranges, and speculative listing spikes. This project constructs an econometric data cleaning funnel and machine learning valuation pipeline that filters noise, benchmarks 6 regression algorithms, and provides real-time fair market property valuations.

Market Dimension	Observed Characteristic	Valuation Impact
Raw Listings Analyzed	13,320 property listings	Spanning over 1,300 unique raw localities in Bangalore
Cleaned Verified Transactions	7,239 properties (241 clusters)	Over 45% statistical and domain noise removed
Primary Price Drivers	Location + Sqft + BHK Configuration	Explains 80.03% of price variance in the city
Champion Model	Ridge Regression (alpha=1.0)	Test $R^2 = 0.8003$ RMSE = 43.08 Lakhs MAE = 17.83 Lakhs
Gradient Boosting Ensemble	MAE = 16.16 Lakhs INR	Lowest absolute error on non-linear price variations
Power BI Integration	powerbi report.pbix Included	Interactive dashboard for spatial market exploration

2. Feature Engineering & Dimensionality Reduction

- Total Square Footage Standardization:** Raw textual ranges (e.g., '2100 - 2850') were parsed to continuous midpoints (2475.0 sq ft). Non-metric unit entries were sanitized.
- BHK & Unit Price Extraction:** Extracted integer bedroom count bhk from description strings and derived normalized unit price: $\text{Price per Sqft} = (\text{Price in Lakhs} \times 100,000) / \text{total_sqft}$.
- Location Dimensionality Reduction:** Localities with ≤ 10 listings were clustered into an 'other' category, reducing sparse dimensions from 1,300+ to **241 high-signal clusters**.

3. Four-Stage Domain & Statistical Outlier Removal

To protect regression algorithms from severe data anomalies, a 4-stage filtering funnel was executed:

- Minimum Area per Bedroom:** Filtered out properties with $< 300 \text{ sq ft/BHK}$ (e.g., 6 BHK in 1000 sq ft).
- Location-Wise Price/Sqft Distribution:** Filtered out speculative listings outside $\mu_L \pm 1.0\sigma_L$ within each locality.
- BHK Price Inversion Filter:** Eliminated anomalies where a 2 BHK was priced higher than a 3 BHK of similar square footage in the same neighborhood.
- Bathroom Ratio Filter:** Removed listings where $\text{bath} > \text{bhk} + 2$.

4. Machine Learning Model Benchmarking (5-Fold Cross-Validation)

Regression Algorithm	5-Fold CV R ²	Holdout Test R ²	RMSE (Lakhs)	MAE (Lakhs)
Decision Tree Regressor (Depth 8)	0.7296	0.6770	54.78 Lakhs	19.88 Lakhs
Random Forest Regressor (100 Trees)	0.7915	0.7088	52.01 Lakhs	17.30 Lakhs
Lasso Regression (alpha=0.1)	0.8038	0.7792	45.29 Lakhs	19.95 Lakhs
Gradient Boosting Regressor	0.8258	0.7427	48.89 Lakhs	16.16 Lakhs
Linear Regression (OLS Baseline)	0.8441	0.7984	43.28 Lakhs	17.84 Lakhs
Ridge Regression (alpha=1.0 - Champion)	0.8401	0.8003	43.08 Lakhs	17.83 Lakhs

5. Interactive Property Valuation Function (`predict\_price`)

An interactive Python valuation engine evaluates one-hot encoded locality vectors alongside spatial metrics:

Locality	Configuration	Total Area	Baths	Estimated Market Value
1st Phase JP Nagar	2 BHK	1,000 sq ft	2	■84.12 Lakhs
1st Phase JP Nagar	3 BHK	1,000 sq ft	3	■86.25 Lakhs
Indira Nagar	2 BHK	1,000 sq ft	2	■181.28 Lakhs
Indira Nagar	3 BHK	1,000 sq ft	3	■183.40 Lakhs
Whitefield	2 BHK	1,200 sq ft	2	■68.50 Lakhs
Electronic City Phase II	2 BHK	1,000 sq ft	2	■51.24 Lakhs

6. Strategic Real Estate Investment Playbook

- **Location Capital Preservation:** Central heritage hubs (Indiranagar, Rajaji Nagar) command 3× higher price per sq ft with low volatility. Allocate capital here for long-term wealth preservation.
- **High-Yield Growth Corridors:** Peripheral IT corridors (Sarjapur, Electronic City) trade at lower entry valuations (■3,500–■5,500/sqft), offering higher gross rental yields (4.5%–6.0%).
- **Developer Inventory Optimization:** Standard 2 BHK and 3 BHK units between 1,000–1,500 sq ft drive over 70% of market absorption. Developers should minimize non-standard atypical floorplans.

7. Technical Conclusion

By enforcing 4-stage domain outlier removal, standardizing complex feature formats, and applying regularized Ridge Regression ($R^2 = 0.8003$), this project provides an automated, statistically grounded valuation engine for Bangalore residential real estate. All executed notebooks and Power BI reports are available in the repository.